Professional Ethical Practice
for Engineering and Computing
(EECS 2911E module 2)

*The word "**ethics**" comes from the Greek word "**ethos**"*
and is defined as:
<u>the study of standards of right and wrong</u>;
part science and part philosophy dealing with
<u>moral conduct, duty, and judgment</u>.

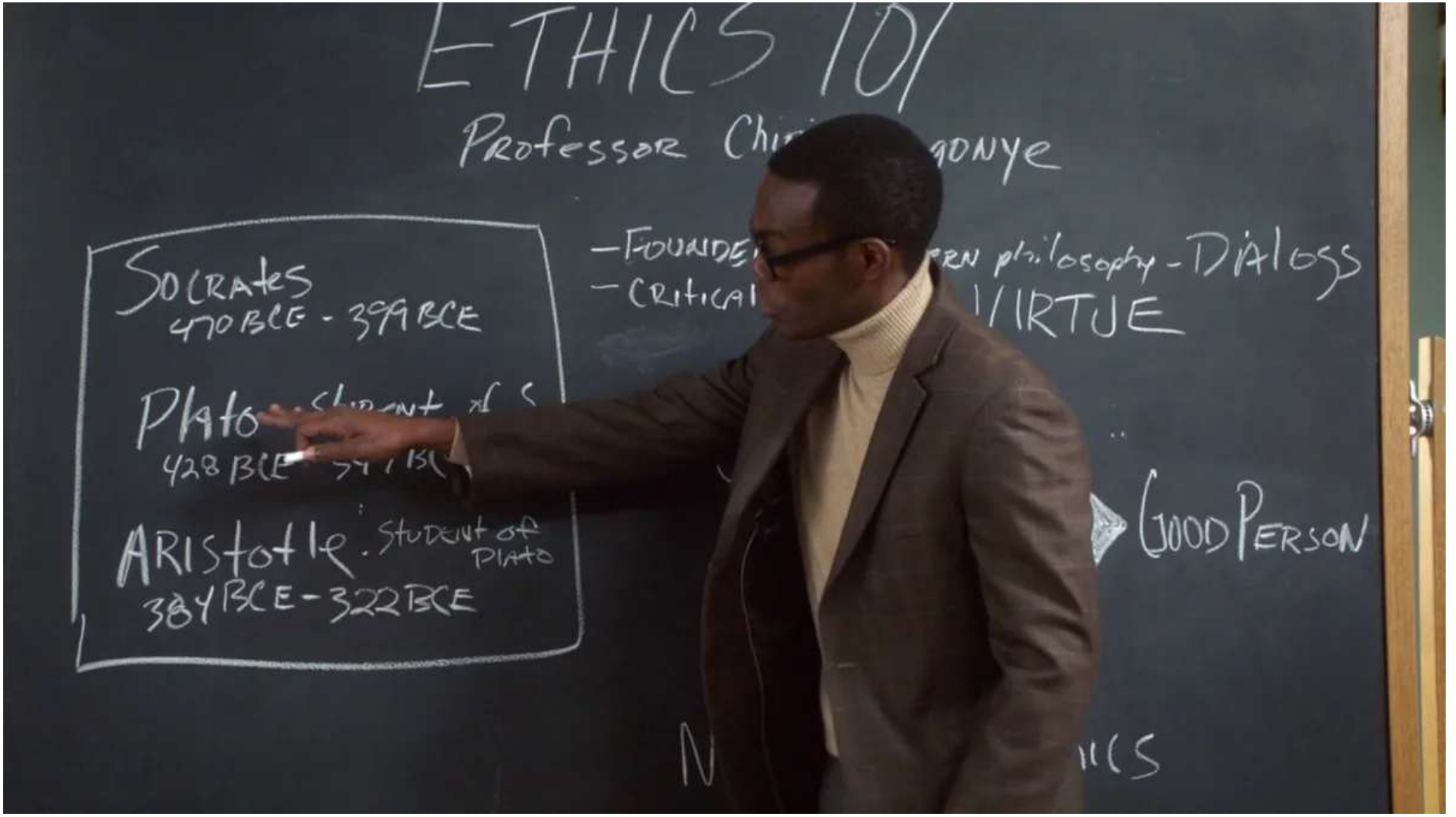
"The unexamined life is not worth living"

..... Socrates

What is Ethics?

Ethics may be *defined* in different ways:

- It refers to well-based standards of **right and wrong** that prescribe what humans **ought** to do, usually in terms of **rights**, **obligations**, benefits to **society**, **fairness**, or specific **virtues**.
- Ethics is also understood as the discipline or field of study dealing with one's ***ethical standard, moral duty or obligation***. This typically gives rise to a ***set of principles or <u>values</u>*** which in turn are used to judge the **appropriateness** of ***particular conduct or behaviour***.



https://www.nbc.com/the-good-place/episodes

Historical Perspectives

Much of what is known about **moral reasoning** mostly began with the ancient Greeks, especially with the philosophers Socrates, Plato, and Aristotle. From the time of ancient Greece up to the current times, there appeared two principal schools of ethics.

- Some moralists maintained that ethical conceptions are inspired in human from above, and they consequently **connected ethics with religion**.
- Other thinkers realized the source of morality is human and maintained that the main **motivational power** of all human actions is found in what is called ***pleasure and happiness,*** and all action is toward this end.

Examples of Levels of Moral reasoning

7. Jenny believes lying is wrong because you could be caught and punished.

8. Joe believes lying is wrong because it can violate the law.

9. Josh believes lying is wrong because it hurts the feelings of others.

10. Jim believes most lying is wrong. But in some case, like saving someone's life, it can be justified.

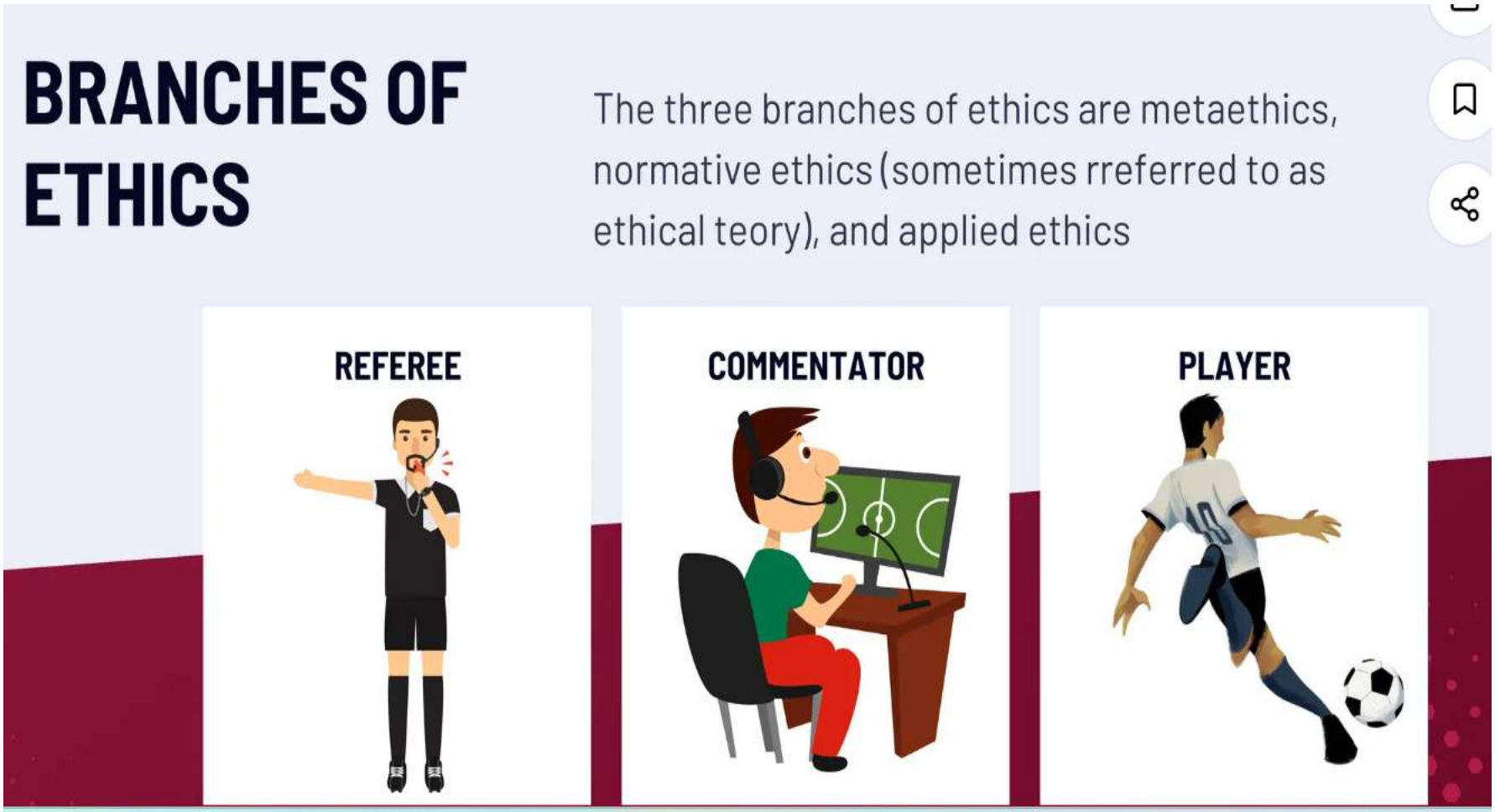
- https://www.chegg.com/

Ethical Theories

- **Metaethics:**
  - What does it mean to say something is ethically **good**?
  - How, if at all, do we know what is **right** and wrong?
  - How do moral attitudes **motivate** action?
  - Are there objective values?
- **Normative Ethics:**
  - **Virtue:** Courage, generosity, compassion.
  - **Deontology:** Act being performed irrespective of consequences.
  - **Consequentialist:** The end justifies the means.

Metaethics talks about the nature of **ethics and moral** reasoning. ..Why?
Normative ethics is interested in determining the content of our **moral** behavior. ..How?
https://axbom.com/digital-ethics-explained/

Sports Analogy



https://www.slideshare.net/slideshow/branches-of-ethics-normative-metaethics-and-applied-ethics/276585107

Ethics Theories

- **Applied ethics**
  - **“ ethics applied to a specific field”**
  - It is a special category of ethical philosophy and the most real of the three categories
  - It deals with tough moral questions and controversial moral issues that people face in their lives including abortion, euthanasia, death penalty, gay/lesbian rights, piracy, etc.
  - May be applied to specific fields like business, environmental, engineering, health, etc.
    - In EECS, we are concerned with public safety, conflicts of interest, client confidentiality, hacking, ransomware, cyber security, cyber-bullying, fake news, A.I. ethics, job loss through automation,….

Modern ethics

Meta-ethics	Normative ethics	Applied ethics	Descriptive ethics
What does "RIGHT" mean?	How should people act?	How do people apply ethical principles?	What do people believe is "right"?
• aka "analytical ethics" • Seeks to understand the nature of ethical concepts themselves	• Deals with the moral "norms" or set of considerations for how people should act	• aka practical ethics • Use of ethical principles to mediate real-life problems	• aka comparative ethics • Deals with what people actually believe (or are made to believe) to be right

https://quizlet.com/gb/771492914/what-is-ethics-jm-020223-flash-cards/

3 Ethical Theories -recap

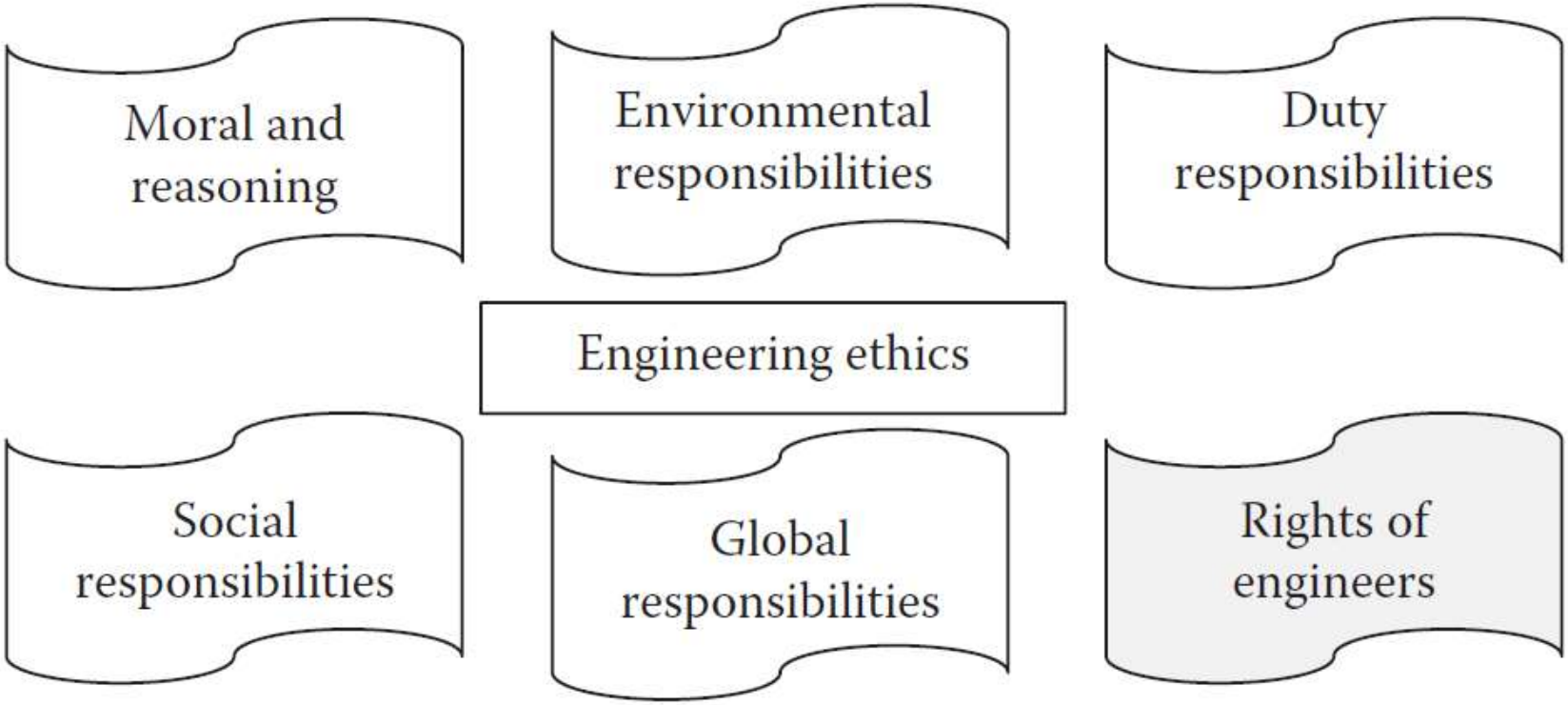
- **Metaethics:**
  - What does it mean to say something is ethically **good**?
    - (what is **good**?)
- **Normative Ethics:**
  - What actions are **good?**
    - **(Is stealing good?)**
- **Applied Ethics**
  - What actions are **good for EECS?**
    - **(Is cyber piracy and hacking good)**

https://www.youtube.com/watch?v=BAswj8evFZk
Ethics by Michael Schur -- Good Place, Office creator
https://youtu.be/BAswj8evFZk

Engineering and Computing Ethics

- Engineering and computing ethics is an area of **applied ethics**, although other theories of ethics come into play in engineering ethics.
- It is the field that **examines and sets standards** for a professional's behaviours and **obligations** to the public, clients, employers, society and the profession.
- Ethical issues involve **quality**, **safety**, legal **compliance**, conflict of **interest** (bribery and gifts), and treatment of **confidential** or proprietary information.

Scopes of Computing and Engineering Ethics



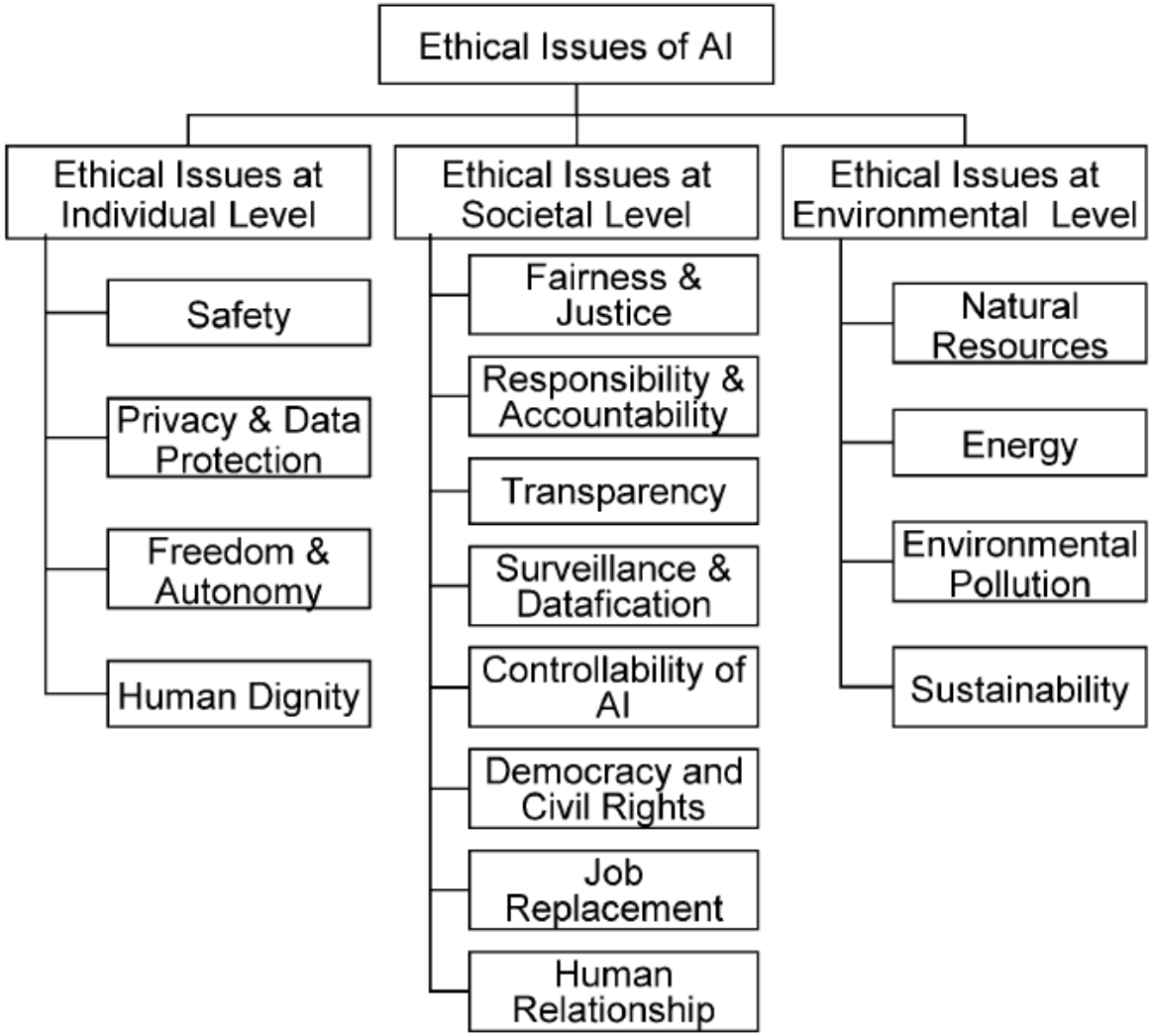
Categories of Computing Ethics

Professional Ethics

- Professional ethics encompasses the **personal**, **organizational**, and **corporate** standard of ***attitude and behavior*** expected of professionals.
- Usually, professional attitude and behavior are guided by "codes of **conduct**," while the operational activities within a profession are guided by "codes of **practice**."
- A professional engineer should apply the code of ethics not in passive performance but as a **set of principles guiding the professional conduct.**

Technology Ethics

- Technology ethics is a transdisciplinary area that draws on theories and techniques from several knowledge domains (such as engineering, science, social sciences, technology, applied ethics, and philosophy) to provide insights into **ethical dimensions of technological systems** and practices for *advancing a technological society*.
- Technology, particularly revolutionary technology, generate many **ethical problems**. Sometimes the problems can be treated easily under extant ethical policies.



https://www.researchgate.net/figure/Branches-of-ethical-theories-91_fig1_362334936

Technological ethical failures

- AI – self learning weapons, robots, drones
    - https://www.machine-ethics.net/machine-ethics-compendium/
    - https://en.m.wikipedia.org/wiki/Machine_ethics
  - Self driving cars – safe? Bug free? Beta test? Ethical choices?
    - https://youtu.be/ixIoDYVfKA0
    - https://www.cnbc.com/2020/01/08/us-safety-agency-opens-probe-into-fatal-tesla-crash-in-indiana.html
    - https://www.extremetech.com/extreme/303538-another-tesla-crash-another-investigation-into-autopilot

- Historical lessons in Technology Ethics – recurring flaw - Hubris
  - Challenger disaster https://www.upi.com/Top_News/US/2016/01/28/Engineer-who-warned-of-1986-Challenger-disaster-still-racked-with-guilt-three-decades-on/4891454032643/
  - Therac 25 automation https://www.computer.org/csdl/magazine/co/2017/11/mco2017110008/13rRUxAStVR
  - 737 MCAS SWhttps://www.seattletimes.com/seattle-news/times-watchdog/the-inside-story-of-mcas-how-boeings-737-max-system-gained-power-and-lost-safeguards/

737 MCAS SW to blame?

[MCAS https://youtu.be/F_7oH4Ozo0k?si=NVthTvm16lxjHvSS](https://youtu.be/F_7oH4Ozo0k?si=NVthTvm16lxjHvSS)

[737 MAX MCAS Ethics Case](https://youtu.be/lb-GCEaYb2c?si=Hpz_JyCQf9UjNg9s)

https://youtu.be/lb-GCEaYb2c?si=Hpz_JyCQf9UjNg9s

5:20 98%

A Boeing 737 Max in Renton, Wash. The company's best-selling model remains grounded after two fatal crashes. Ruth Fremson/The New York Times

By Natalie Kitroeff, David Gelles and Jack Nicas

Published Oct. 2, 2019 Updated Oct. 29, 2019

A senior Boeing engineer filed an internal ethics complaint this year saying that during the development of the 737 Max jet the company had rejected a safety system to minimize costs, equipment that he felt could have reduced risks that contributed to two fatal crashes.

Read this story in the app

5:16 98%

mcas software 737 blame

Transport Canada official says software blamed for Boeing Max 8 crashes must ...

Nov. 25, 2019 · In this May 8, 2019, file photo, a Boeing 737 MAX 8 jetliner being built for ... The MCAS software, which automatically pushes the nose of the aircraft down ...

https://www.theregister.co.uk › ...

'Software delivered to Boeing' now blamed for 737 Max warning fiasco ...

May 7, 2019 · 'Software delivered to Boeing' now blamed for 737 Max warning fiasco ... The AOA sensors feed the controversial MCAS trim system, another software ...

BI https://www.businessinsider.com › ...

Boeing CEO explains why 737 Max pilots were not told of MCAS software ...

Apr. 29, 2019 · Boeing CEO Dennis Muilenburg explained on Monday why Boeing

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Software Errors Caused deaths?

- In the early 1980s, Atomic Energy of Canada Limited made the Therac-25 cancer radiation treatment machine
- Between 1985 and 1987, radiation overdoses by the Therac-25 caused severe burns, which killed three patients and seriously injured three others

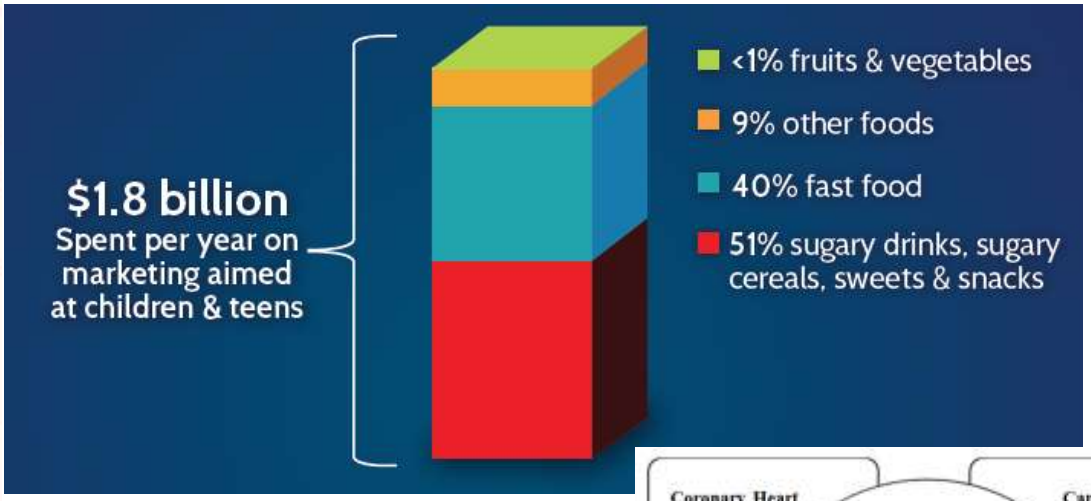
- Conditions for moral responsibility
  - Causal condition: actions (or inactions) caused the harm
  - Mental condition
    - Actions (or inactions) intended or willed -OR-
    - Moral agent is careless, reckless, or negligent
- Therac-25 team morally responsible
  - They constructed the device that caused the harm
  - They were negligent

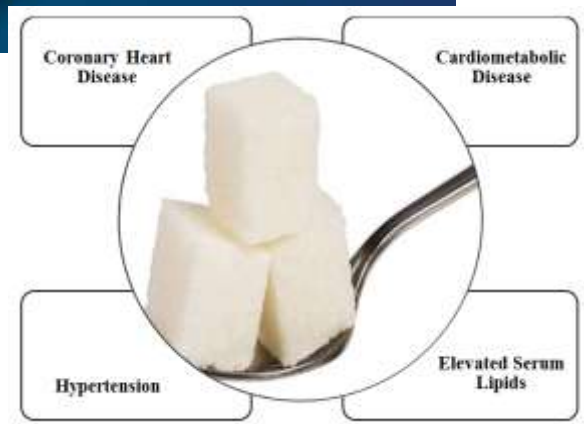
"groupthink"

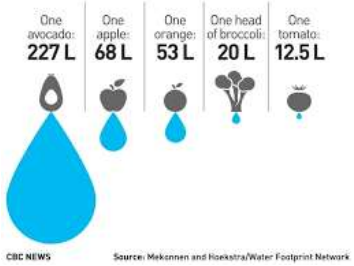
Social Ethics

- Social ethics deals with **human needs** and aspirations.
  - involves a **code of conduct** created by a **society** in order to ensure a smooth functioning of the said society.
- **Engineering** has enormous role to help provide **benefits to society** and the idea of **<u>social responsibility</u>** is very common in engineering ethics.
  - helps to provide **basic needs** such as water, food, housing, and energy, and does that on a way crucial for industrial sector to function.

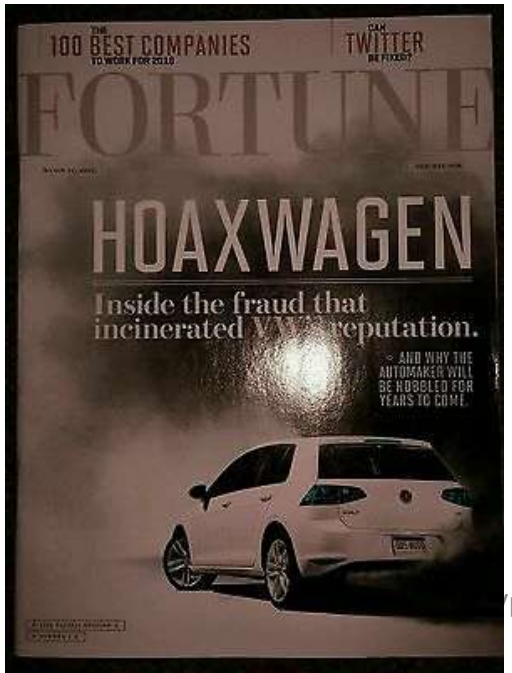
Food Industry Ethical Lapses –How to help?







Auto Industry Ethical Lapses



/ELG/S

Professional Regulations and By-Laws

The Act specifies the required Regulations and By-laws and the Association defines the details.

- **Regulations**
  - These provide the specific rules for engineers
  - For example:
    - the Act states that academic and experience requirements are needed for licensing
    - Association sets these requirements in detail
- **By-Laws**
  - These provide specific rules for running the Association
  - For example:
    - Frequency of council meetings, number of members on the discipline committee and complaints committee, etc.

Self Regulation

- Professions that have been given the right and responsibility of self-regulation, including the engineering profession, have tended to opt for the first alternative, espousing sets of underlying principles as codes of professional ethics which form the basis and framework for responsible professional practice.
- Arising from this context, professional codes of ethics have sometimes been incorrectly interpreted as a set of “rules” of conduct intended for passive observance.
- Therefore, a code of professional ethics is more than a minimum standard of conduct; rather, it is a set of principles which should guide professionals in their daily work.

Code of Conducts and Ethics

- A code of ethics is the characteristic of a profession. It is an **aspirational** system of principles and rules or the guideline that sets satisfactory behaviors for a given group of people or profession.
- It makes obvious what the public anticipates from the profession. It expresses the rights, duties, and obligations of the members of the profession.
- Professionally, the code of ethics is a comprehensive guide to professional conduct. It is usually designed to help practitioners maintain the highest level of ethical conduct, standards of practice and integrity with respect to their professional activities.

The Beginning of Code of Ethics

- In the mid-twentieth century there was a proliferation of important ethics codes that still guide engineering professional behavior and research activities.
- The first reference to a National Society of Professional Engineers (NSPE) code of ethics is found in the May 1935 issue of *The American Engineer* in the form of a suggestion for membership consideration.
- In 1946, the Board approved the canons of ethics for engineers as prepared by a joint committee sponsored by the Engineers' Council for Professional Development, a coordinating body of technical engineering societies.

In Canada

- Engineers Canada is the national organization of the 12 provincial and territorial associations that regulate the profession of engineering in Canada. Established in 1936, Engineers Canada serves the associations, which are its constituent and sole members, through the delivery of national programs which ensure the highest standards of engineering education, professional qualifications and ethical conduct.
- Provincial and territorial associations of professional engineers are responsible for the regulation of the practice of engineering in Canada. Each association has been established under a professional engineering Act of its provincial or territorial legislature and serves as the regulatory body for the practice of engineering within its jurisdiction.

Engineers Canada

- Engineers Canada is the national organization of the provincial and territorial associations that regulate the practice of engineering in Canada and license the country's 290,000 members of the engineering profession.
- Engineering is a self-regulated profession. Engineers Canada exists to support the provincial and territorial engineering regulatory bodies. Together, we work to advance the profession in the public interest.
- Engineers Canada has national position statements on key issues relating to the public interest, including infrastructure, labor mobility and regulating the profession. These are consensus positions of the regulatory bodies.

Engineers Canada

- **Standards**
  - accredits undergraduate engineering programs
  - develops professional practice and qualifications guidelines as it relates to the public interest
  - facilitates international and interprovincial labour mobility
  - owns the official marks on the terms ENGINEER, ENGINEERING, PROFESSIONAL ENGINEER, P.ENG., CONSULTING ENGINEER, INGÉNIEUR, ING., INGÉNIEUR CONSEIL, GÉNIE and INGÉNIERIE
- **Outreach**
  - promotes the engineering profession
  - conducts federal government relations
  - develops policies and positions
  - provides insurance, financial and other benefits
  - undertakes national media relations
- **Research**
  - monitors university enrollment
  - promotes diversity in the engineering profession
  - tracks labor market trends

General Interpretation of Code of Ethics

- **Protection of the Public and the Environment**
  - Safety, health and welfare of the public and the protection of the environment.
- **Faithful Agents of Client and Employers**
  - Engineers shall act as faithful agents or trustees of their clients and employers with objectivity, fairness, and justice to all parties.
- **Competence and Knowledge**
  - Professional engineers shall offer services, advise on, or undertake engineering assignments only in areas of their competence by virtue of their training and experience.
- **Fairness and Integrity in the Workplace**
  - Honesty, integrity, competence, devotion to service, and dedication to generally enhancing the quality of life are cornerstones of professional responsibility.
- **Professional Accountability and Leadership**
  - Engineers have a duty to practice in a careful and diligent manner and accept responsibility and be accountable for their actions.

Professional Engineers Ontario

Regulation R.R.O. 1990, Reg. 941

- The set of regulations created by Professional Engineers Ontario under the authority of the Professional Engineers Act
  - https://www.ontario.ca/laws/regulation/900941
- Contains the Regulations and the By-Laws for Ontario
- Key Sections:
  - Section 72 Professional Misconduct
    - Defines Professional Misconduct
  - Section 77 Code of Ethics
    - Rules of personal conduct to guide the engineer
  - We will look at these in detail in this lecture.

Professional Engineers Ontario

Definition of Professional Misconduct

- **72.** (1) In this section,

  "harassment" means engaging in a course of vexatious comment or conduct that is known or ought reasonably to be known as unwelcome and that might reasonably be regarded as interfering in a professional engineering relationship;

  "negligence" means an act or an omission in the carrying out of the work of a practitioner that constitutes a failure to maintain the standards that a reasonable and prudent practitioner would maintain in the circumstances. R.R.O. 1990, Reg. 941, s. 72 (1); O. Reg. 657/00, s. 1 (1).

Professional Engineers Ontario

Definition of Professional Misconduct

- 72.(2) For the purposes of the Act and this Regulation, **"professional misconduct"** means,

  (a) negligence,

  (b) failure to make reasonable provision for the safeguarding of life, health or property of a person who may be affected by the work for which the practitioner is responsible,

  (c) failure to act to correct or report a situation that the practitioner believes may endanger the safety or the welfare of the public,

  (d) failure to make responsible provision for complying with applicable statutes, regulations, standards, codes, by-laws and rules in connection with work being undertaken by or under the responsibility of the practitioner,

Professional Engineers Ontario

Definition of Professional Misconduct

(e) signing or sealing a final drawing, specification, plan, report or other document not actually prepared or checked by the practitioner,

(f) failure of a practitioner to present clearly to the practitioner's employer the consequences to be expected from a deviation proposed in work, if the professional engineering judgment of the practitioner is overruled by non-technical authority in cases where the practitioner is responsible for the technical adequacy of professional engineering work,

(g) breach of the Act or regulations, other than an action that is solely a breach of the code of ethics,

(h) undertaking work the practitioner is not competent to perform by virtue of the practitioner's training and experience,

Definition of Professional Misconduct

(i) failure to make prompt, voluntary and complete disclosure of an interest, direct or indirect, that might in any way be, or be construed as, prejudicial to the professional judgment of the practitioner in rendering service to the public, to an employer or to a client, and in particular, without limiting the generality of the foregoing, carrying out any of the following acts without making such a prior disclosure:

- Accepting compensation in any form for a particular service from more than one party.
- Submitting a tender or acting as a contractor in respect of work upon which the practitioner may be performing as a professional engineer.
- Participating in the supply of material or equipment to be used by the employer or client of the practitioner.
- Contracting in the practitioner's own right to perform professional engineering services for other than the practitioner's employer.
- Expressing opinions or making statements concerning matters within the practice of professional engineering of public interest where the opinions or statements are inspired or paid for by other interests,

Definition of Professional Misconduct

(j) conduct or an act relevant to the practice of professional engineering that, having regard to all the circumstances, would reasonably be regarded by the engineering profession as disgraceful, dishonourable or unprofessional,

(k) failure by a practitioner to abide by the terms, conditions or limitations of the practitioner's licence, provisional licence, limited licence, temporary licence or certificate,

(l) failure to supply documents or information requested by an investigator acting under section 33 of the Act,

(m) permitting, counselling or assisting a person who is not a practitioner to engage in the practice of professional engineering except as provided for in the Act or the regulations,

(n) harassment.

R.R.O. 1990, Reg. 941, s. 72 (2); O. Reg. 657/00, s. 1 (2); O. Reg. 13/03, s. 19.

PEO Code of Ethics

Stakeholders

Professional Engineers Ontario

- 77.1 It is the duty of a practitioner
  - to the public
  - to the practitioner's employer
  - to the practitioner's clients
  - to other members of the practitioner's profession
  - and to the practitioner
- to act at all times with
  - …

PEO Code of Ethics

Core Duties

- 77.1. (continued) to act at all times with:
  - i. fairness and loyalty to the practitioner's associates, employers, clients, subordinates and employees;
  - ii. fidelity to public needs;
  - iii. devotion to high ideals of personal honour and professional integrity;
  - iv. knowledge of developments in the area of professional engineering relevant to any services that are undertaken; and
  - v. competence in the performance of any professional engineering services that are undertaken.

PEO Code of Ethics

Duty to Public

Professional Engineers Ontario

- 77.2. A practitioner shall,

  i. regard the practitioner's duty to public welfare as paramount;

  ii. endeavour at all times to enhance the public regard for the practitioner's profession by

  - extending the public knowledge thereof and
  - discouraging untrue, unfair or exaggerated statements with respect to professional engineering;

PEO Code of Ethics

Truthfulness and Transparency

- 77.2. (continued) A practitioner shall

  iii. Not express publicly, or while the practitioner is serving as a witness before a court, commission or other tribunal, opinions on professional engineering matters that are not founded on adequate knowledge and honest conviction;

  iv. Endeavor to keep the practitioner's licence, temporary licence, limited licence or certificate of authorization, as the case may be, permanently displayed in the practitioner's place of business.

PEO Code of Ethics

Duties to Employer

Professional Engineers Ontario

- 77.3. A practitioner shall
  - act in professional engineering matters for practitioner's employer as a faithful agent or trustee and shall
  - regard as confidential information obtained by the practitioner as to the business affairs, technical methods or processes of an employer and
  - avoid or disclose a conflict of interest that might influence the practitioner's actions or judgment.

PEO Code of Ethics

Conflict of Interest

Professional Engineers Ontario

- 77.4. A practitioner must disclose immediately to the practitioner's client any interest, direct or indirect, that might be construed as prejudicial in any way to the professional judgment of the practitioner in rendering service to the client.

PEO Code of Ethics

Moonlighting

Professional Engineers Ontario

- 77.5. A practitioner who is an employee-engineer *and* is contracting in the practitioner's own name to perform professional engineering work for other than the practitioner's employer, must
  - provide the practitioner's client with a written statement of the nature of the practitioner's status as an employee and the attendant limitations on the practitioner's services to the client, must
  - satisfy the practitioner’s duty to the practitioner’s employer, and must
  - inform the practitioner’s employer of the work.

PEO Code of Ethics

Cooperation

Professional Engineers Ontario

- 77.6. A practitioner must co-operate in working with other professionals engaged on a project.

PEO Code of Ethics

Duty to other Engineers

Professional Engineers Ontario

- 77.7. A practitioner shall,
  i. act towards other practitioners with courtesy and good faith,
  ii. not accept an engagement to review the work of another practitioner for the same employer except with the knowledge of the other practitioner or except where the connection of the other practitioner with the work has been terminated,
  iii. not maliciously injure the reputation or business of another practitioner;

PEO Code of Ethics

Duty to Other Engineers

Professional Engineers Ontario

- 77.7 (continued) A practitioner shall,

  iv. not attempt to gain an advantage over other practitioners by paying or accepting a commission in securing professional engineering work, and

  v. give proper credit for engineering work,

  uphold the principle of adequate compensation for engineering work,

  provide opportunity for professional development and advancement of the practitioner's associates and subordinates, and

  extend the effectiveness of the profession through the interchange of engineering information and experience.

PEO Code of Ethics

Honour and Duty to Report

Professional Engineers Ontario

- 77.8. A practitioner shall maintain the honour and integrity of the practitioner's profession and without fear or favour expose before the proper tribunals unprofessional, dishonest or unethical conduct by any other practitioner.
- R.R.O. 1990, Reg. 941, s. 77; O. Reg. 48/92, s. 1.

- http://peo.on.ca/index.php?ci_id=1815&la_id=1
  - (ethics of whistleblowing?)

Ethics versus Misconduct

- A Code of Ethics defines lines you should not cross
- Misconduct define the lines you must not cross.
- In general, Codes of Ethics go beyond what is required of the law
  - Conversely, just because something is not illegal does not make it right
- Respecting the Code of Ethics protects you
  - If you respect the code of ethics you are free of misconduct

Ethics versus Misconduct

- This table shows the relationships between the code of ethics (R77) and the definition of professional misconduct (R72)
- Every case of misconduct breaches at least one section of the CoE (DISCLAIMER: this is ***not legal advice***! Do the right thing!)

The Code of Ethics protects you from Professional Misconduct
In general, if you respect every clause of the code of ethics you are very unlikely to commit professional misconduct.
Conversely, every case of professional misconduct breaches at least one clause of the code of ethics.
This table shows the relationships between the CoE and Prof. Misconduct.
Cells with a "1" are the strongest relationships. Secondary relationships are shown with a "2"

		Professional Misconduct													
	Code of Ethics	R72.2(a) Negligence	R72.2(b) Protect Public	R72.2(c) Duty to Report	R72.2(d) Breach of Standards	R72.2(e) Misrepresentation	R72.2(f) Duty to warn	R72.2(g) Breach of Act	R72.2(h) Incompetence	R72.2(i) Conflict of Interest	R72.2(j) Dishonourable Conduct	R72.2(k) Breach of License Terms	R72.2(l) Obstruction	R72.2(m) Aiding and Abetting	R72.2(n) Harassment
R77.1.i	Fairness and Loyalty to all						1			2	2				1
R77.ii	Fidelity to public needs		1	2							2				
R77.1.iii	Honor and Prof Integrity					1	2	1		2	1	1	1	1	2
R77.1.iv	Continuing Education	2			1										
R77.1.v	Competence	1							1		2				
R77.2.i	Duty to public #1			2							2				
R77.2.ii	Promote Profession														
R77.2.iii	Honest and informed testimony										2				
R77.2.iv	Display license														
R77.3	Duties to Employer Faithful, Confidential, Conflict of interest						2			2	2				
R77.4	Conflict of Interest									1	2				
R77.5	Moonlighting									2	2				
R77.6	Cooperate										2				2
R77.7.i	Duty to Colleagues Courtesy and Good Faith										2		2		2
R77.7.ii	Duty to Colleagues Disclose review					2					2				2
R77.7.iii	Duty to Colleagues Reputation										2				2
R77.7.iv	Duty to Colleagues secret commissions									2	2				
R77.7.v	Duty to Colleagues Credit, compensation, help, share					2					2				
R77.8	Duty to Report			1							2		2		

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Cells with a "1" are the strongest relationships. Secondary relationships are shown with a "2"

		Professional Misconduct													
		R72.2(a)	R72.2(b)	R72.2(c)	R72.2(d)	R72.2(e)	R72.2(f)	R72.2(g)	R72.2(h)	R72.2(i)	R72.2(j)	R72.2(k)	R72.2(l)	R72.2(m)	R72.2(n)
	Code of Ethics	Negligence	Protect Public	Duty to Report	Breach of Standards	Misrepresentation	Duty to warn	Breach of Act	Incompetence	Conflict of Interest	Dishonourable Conduct	Breach of License Terms	Obstruction	Aiding and Abetting	Harassment
R77.1.i	Fairness and Loyalty to all						1			2	2				1
R77.ii	Fidelity to public needs		1	2							2				
R77.1.iii	Honor and Prof Integrity					1	2	1		2	1	1	1	1	2
R77.1.iv	Continuing Education	2			1										
R77.1.v	Competence	1							1		2				
R77.2.i	Duty to public #1			2							2				
R77.2.ii	Promote Profession														
R77.2.iii	Honest and informed testimony										2				
R77.2.iv	Display license														
R77.3	Duties to Employer Faithful, Confidential, Conflict of interest						2			2	2				
R77.4	Conflict of Interest									1	2				
R77.5	Moonlighting									2	2				
R77.6	Cooperate										2				2
R77.7.i	Duty to Colleagues Courtesy and Good Faith										2		2		2
R77.7.ii	Duty to Colleagues Disclose review					2					2				2
R77.7.iii	Duty to Colleagues Reputation										2				2
R77.7.iv	Duty to Colleagues secret commissions									2	2				
R77.7.v	Duty to Colleagues Credit, compensation, help, share					2					2				
R77.8	Duty to Report			1							2		2		

Ethical Decision Making

- Ethical decision making is not always easy.
- Ethical Dilemmas, where one needs to balance his/her duty to multiple stakeholders is common.
  - Remember: Public safety is paramount (R77.2.i)
- The key is to be systematic
- There are two scenarios for ethical decision making:
  - You need to determine a course of action
  - You need to decide if a course of action taken was ethical and/or misconduct.

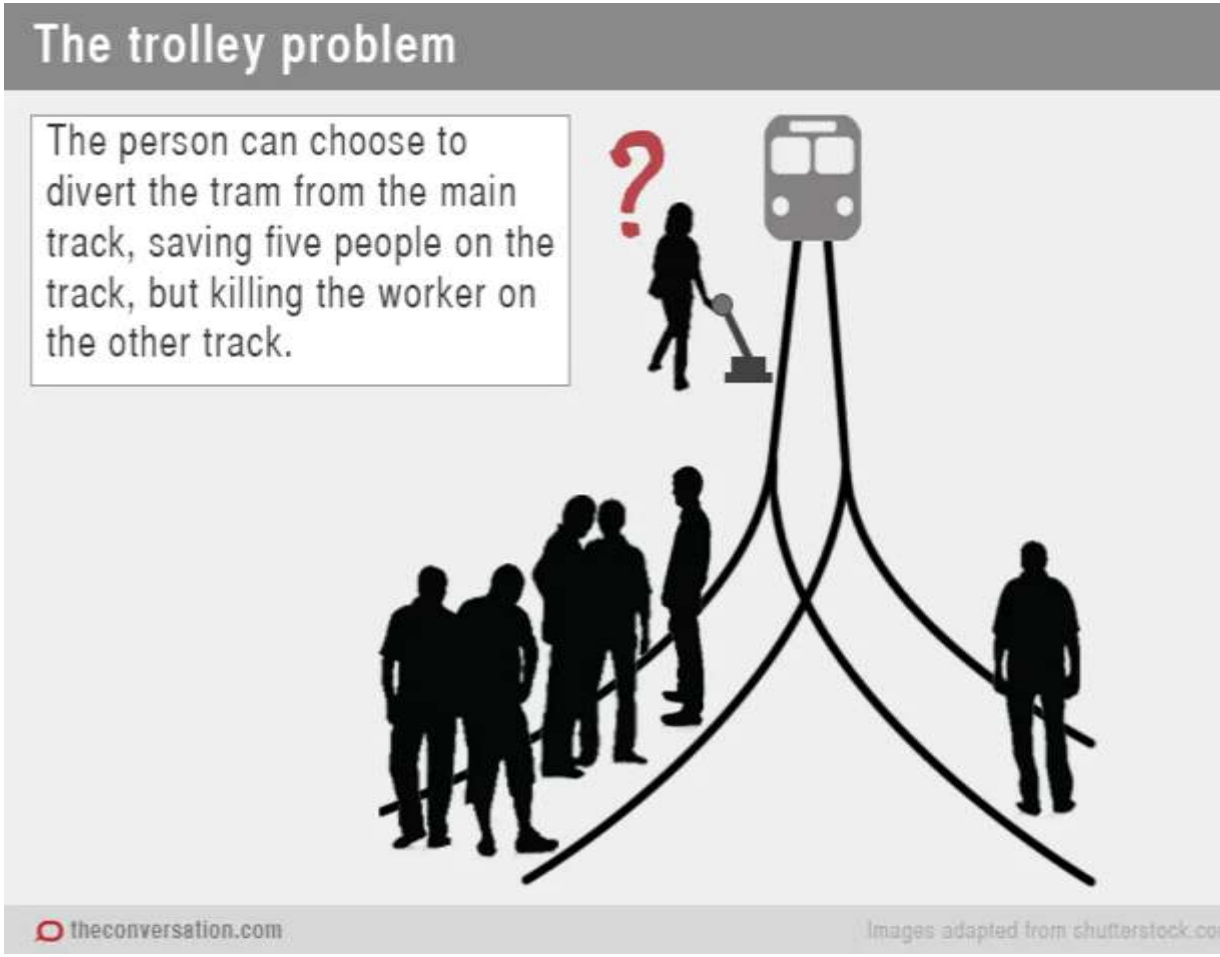
Ethical Decision Making

- If you need to determine a course of action:
  - Get the facts straight
  - Develop optional courses of action
    - “Do nothing” is an option
  - Consider consequences of each course of action
    - Both Ethical and Professional Conduct
    - Clearly outline your logic (reference sections of Regs)
  - Chose the best option
    - Identify potential impacts of the selected course of action

Ethical Decision Making

- If you need to decide if ethics/conduct were breached:
  - Read and understand the case
    - Get the facts straight
    - Clearly list your assumptions
  - Identify the sections of the code of ethics that are relevant
  - Identify sections of the definition of misconduct that apply
  - Clearly outline your decision (reference sections of Regs)

Difficult Ethical Decisions



https://youtu.be/ixIoDYVfKA0?si=rn0g3T9o-JXmTDEa
Self driving ethics

Case Study

Duty to Report (R77.8,R72.2.c)

- If engineer observes unsafe, unethical or illegal actions
  - Code of ethics requires that action be taken.
    - May be necessary to “blow the whistle”
      - Get the facts
      - Consider possible solutions
      - Try to resolve issue with responsible person
      - Escalate internally if necessary
      - Contact the Association for mediation and guidance if necessary
      - Only “go public” as a last resort
    - Failure to report is Professional Misconduct (R.R.O. 1990, Reg. 941 section 72.2(c))
    - PEO provides guidance on reporting:

https://www.peo.on.ca/public-protection/complaints-and-unlicensed-practice/report-unlicensed-individuals-or-companies#:~:text=Contact%20us%20about%20Unlicensed%20Practice,1.800.339.3716

Case Study

Duty to Report (R77.8,R72.2.c)

- EXAMPLE: Engineer learns employer is dumping toxic waste into river
  - This is violation of Environmental Protection Act (illegal)
  - What should the engineer do?

Case Study

Use of Computer Programs

- Ethical concerns created by use of software
  - Liability of computer errors
  - Software piracy
  - Plagiarism from Internet
  - Protection against computer viruses

    - https://ethicsinaction.ieee.org/

Case Study

Liability of Computer Errors

- Engineering failure results from faulty software
  - Engineer uses and bases decision on software output.
  - Incorrect or misunderstood output results in failure.
- Engineer is responsible for failure
- When software used in engineering decisions engineer should
  - Be competent in area in which software is used
  - Know what assistance software is providing
  - Know theory and assumptions used by software
  - Know range and limits of its validity
  - Test software to ensure accuracy.

Software Piracy

- An Engineer borrows a software licence key from a friend to perform some calculations for work.
- Unauthorized use or copying is theft
- Software protected by copyright / trademark laws.
- Canadian Alliance Against Software Theft / Business Software Alliance detects and prosecutes copyright infringement.
- Ethical use of software
  - Stimulates software development
  - Increases creativity, productivity and job opportunities.

Case Study

The Engineer's Seal

- An engineer is asked by his employer to seal an engineering document.
- The employer says the deadline is in 2 days and the engineer that prepared the document is on vacation.
- What do you do?

Seal must be signed and dated to be valid

Duty of the Association

Enforcement and Discipline

- Duty of associations is to protect public
  - Respond to complaints from public.
  - Association investigates, mediates, takes legal action if necessary.
- Unlicensed: Enforcement
- Licensed: Discipline

Enforcement and Discipline

- Enforcement of Act
  - Against law to practise without license.
  - Association enforces law by prosecuting offenders in court.
- Professional Discipline
  - Licensed member can be disciplined for misconduct, incompetence, unethical behavior and other reasons
  - Fines, revocation or suspension of licence, restriction of work.

Enforcement

- Enforcement: legal action PEO takes against individuals or entities who practise engineering without a licence, or offer engineering services to the public without a Certificate of Authorization.
- Sections 39 and 40 of the Professional Engineers Act give PEO the authority to take action.
- Section 40(1) sets maximum fines for individuals and firms practising without the necessary licences (P.Eng., and/or C of A). Fines are $25,000 for a first offence and $50,000 for each subsequent offence.
- Section 40(2) authorizes fines of up to $10,000 for a first offence and $25,000 for each subsequent offence, for people who:
  - use the title "professional engineer" or an abbreviation or variation as an occupational designation;
  - use a term, title or description that will lead to the belief that the person may engage in the practice of professional engineering; or
  - use a seal that will lead to the belief that the person is a professional engineer.

Discipline

- Association staff receive and administer the complaints
- Discipline decisions are made by a discipline committee composed of engineers.
- Main reasons for disciplinary action
  - Professional misconduct
  - Incompetence
  - Breach of the code of ethics
  - Physical of mental incapacity
  - Conviction of a serious offence

Complaints Committee

Professional Engineers Ontario

- Complaints Committee from R.S.O. 24(1)
- http://www.e-laws.gov.on.ca/html/statutes/english/elaws_statutes_90p28_e.htm#BK24

- The Complaints Committee shall consider and investigate complaints made by members of the public or members of the Association regarding the conduct or actions of a member of the Association or holder of a certificate of authorization, a temporary licence, a provisional licence or a limited licence
  - Determine whether or not to forward complaint to discipline committee

Common Professional Complaints (1 of 2)

- Conflict of Interest
  - Personal preference or financial interest which interferes with duty to employer.
    - Payment from more than one source for same service
    - Secret commission
  - Must avoid conflicts of interest
  - If cannot be avoided, must provide prompt, voluntary and complete (PVC) disclosure to all involved

Common Professional Complaints (2 of 2)

- Breach of standards
  - Design work: Engineer must follow commonly accepted codes and standards.
  - First step in design process: Information gathering
  - Start of project: Engineer should find appropriate design standards / safety regulations.

Discipline Committee

Professional Engineers Ontario

- Discipline Committee from R.S.O. 28(1) – 28(4)
  http://www.e-laws.gov.on.ca/html/statutes/english/elaws_statutes_90p28_e.htm#BK28
  - Duty to:
    - hear and determine allegations of professional misconduct or incompetence of a member
  - Power to:
    - Revoke a license
    - Suspend a license
    - Impose terms, conditions or limitations including successful completion of a particular course or course of study
    - Fine of up to $5000
    - Publicly shame
      - (publish in Dimensions)

Summary of the CIPS Code of Ethics and Professional Conduct

http://www.cips.ca/ethics

- CIPS members must protect the public interest and discharge with integrity all duties and services owed to:
  - the public
  - CIPS members
  - other IT professionals, and
  - clients.
- The obligation to protect the public interest is paramount
- Integrity and trustworthiness are fundamental and are key elements of each requirement of the Code or Ethics
- Dishonourable or questionable conduct will result in CIPS taking disciplinary action.

Summary of the CIPS Code of Ethics and Professional Conduct

http://www.cips.ca/ethics

1. **Protect the Public Interest and Maintain Integrity**
   - Work with due regard for public health, public safety and the environment
   - Uphold the reputation of CIPS and the profession
   - Report problems that may injure persons, organizations, property or the economy
   - Not discriminate on any grounds, such as race, sex, sexual orientation, nationality, social origin, family status or disability

Summary of the CIPS Code of Ethics and Professional Conduct

2. Demonstrate Competence and Quality of Service

- Exercise uncompromised professional judgment
- Be honest and candid when providing service
- Maintain professional competency and remain current
- Be aware of and compliant with legislation, standards and bodies of knowledge
- Respect rights of third parties, such as giving credit where it is due
- Conduct all activities with due regard for ownership, property rights and privacy.

Summary of the CIPS Code of Ethics and Professional Conduct

3. Maintain Confidential Information and Privacy

- Do not disclose having been retained by a client unless the nature of the matter requires such disclosure or the client has expressly given permission to disclose such information
- Do not repeat gossip about a client's business or affairs
- Treat all client business information as confidential, and respect copyrights, trade secrets, privacy and terms of licence or applicable agreements.
- Understand and comply with any obligations imposed under privacy legislation, including *The Personal Information Protection and Electronics Document* (PIPEDA) privacy Act and other laws.

Summary of the CIPS Code of Ethics and Professional Conduct

4. Avoid Conflicts of Interest

- Do not place personal or professional interests or those of colleagues above the interest of the public or those of clients
- Make reasonable efforts to notify all parties involved to make full disclosure if any conflict is seen to occur by an independent third party
- Do not create hidden reciprocal relationships with third parties who stand to gain as a result of their work
- Inform clients of any interest which might be seen to impair professional judgement.

Summary of the CIPS Code of Ethics and Professional Conduct

5. Uphold Responsibility to the IT Profession

- Participate and act with integrity in a manner that upholds the reputation and good standing of CIPS and the profession
- Participate with and support fellow CIPS members in their professional development, and provide opportunities for the professional development of new CIPS members, potential CIPS members, colleagues and subordinates
- Make reasonable efforts to volunteer time and expertise by participating in activities to promote the advancement of CIPS and the profession
- Exemplify the values of equality, tolerance and respect for others.

CIPS Code of Ethics

Case Study 1

- Your company has developed safety critical software that you believe may not have been adequately tested
- You mention this to your manager, and he tells you "you are too inexperienced to make that judgment, the expert testers are confident the software is OK".
- What do you do?

CIPS Code of Ethics

Case Study: Possible Answer

1. Get the facts straight

- Do you have proof that the software was not tested properly?
- Can you propose the tests that should be done and estimate the cost?
- Do the expert testers agree with your bosses claim of adequate testing?

2. Develop optional courses of action

- Do nothing
- Try to convince your boss that more testing is needed
- Try to convince the testers that more testing is needed
- Go public
- Report to CIPS

CIPS Code of Ethics

Case Study 1: Possible Answer (Continued)

3. Consider consequences of each course of action

- Protect the Public Interest and Maintain Integrity
  - Work with due regard to public safety (public safety is paramount)
  - Uphold the reputation of CIPS and the Profession (don’t go public)
  - Duty to Report
- Demonstrate Competence and Quality of Service
  - Exercise uncompromised professional judgment (validate the facts, speak to testers)
- Maintain Confidential Information
  - Don’t repeat gossip (get the facts)
  - Client info confidential (don’t go public)

CIPS Code of Ethics

Case Study 1: Possible Answer (Continued)

4. Select Best Option (or combination)

- Validate the fact that the software has not been tested adequately. Talk to the testers. They may validate the boss' statement (testing adequate) or agree with you.
- If more testing is needed (testers agree) then discuss with your boss (give facts, solution, duty to report).
- If boss still refuses more test, escalate internally (give facts, solution, duty to report).
- If there is still no will to correct the deficiency, report to CIPS for guidance.

CIPS Code of Ethics

Case Study 2

- You know how to crack the encryption on cell phone calls and have the hardware and software available to do this.

- Is there any ethical situation when you might consider it ‘right’ to use this knowledge.

IEEE Code of Ethics

IEEE is the trusted voice for engineering, computing and technology information around the globe

https://www.ieee.org/about/corporate/governance/p7-8.html

1. to hold paramount the safety, health, and welfare of the public, to strive to comply with ethical design and sustainable development practices, and to disclose promptly factors that might endanger the public or the environment;
2. to avoid real or perceived conflicts of interest whenever possible, and to disclose them to affected parties when they do exist;
3. to be honest and realistic in stating claims or estimates based on available data;
4. to reject bribery in all its forms;
5. to improve the understanding by individuals and society of the capabilities and societal implications of conventional and emerging technologies, ….;
6. to maintain and improve our technical competence and to undertake technological tasks for others only if qualified by training or experience, or after full disclosure of pertinent limitations;
7. to seek, accept, and offer honest criticism of technical work, to acknowledge and correct errors, and to credit properly the contributions of others;
8. to treat fairly all persons and to not engage in acts of discrimination based on race, religion, gender, disability, age, national origin, sexual orientation, gender identity, or gender expression;
9. to avoid injuring others, their property, reputation, or employment …..
10. to assist colleagues and co-workers in their professional development and to support them in following this code of ethics.

The ACM (Association for Computing Machinery) focuses on computer science theory, software, and applications and has 100,000 members.

ACM Code of Ethics – 1 of 2

See https://www.acm.org/binaries/content/assets/about/acm-code-of-ethics-booklet.pdf

Short version:

1. Software engineers shall act consistently with the public interest.
2. Software engineers shall act in a manner that is in the best interests of their client and employer *consistent with the public interest*.
3. Software engineers shall ensure that their products and related modifications meet the highest professional standards possible.
4. Software engineers shall maintain integrity and independence in their professional judgment.

ACM Code of Ethics – 2 of 2

Short version continued:

5. Software engineering managers and leaders shall subscribe to and promote an ethical approach to the management of software development and maintenance.
6. Software engineers shall advance the integrity and reputation of the profession consistent with the public interest.
7. Software engineers shall be fair to and supportive of their colleagues.
8. Software engineers shall participate in lifelong learning regarding the practice of their profession and shall promote an ethical approach to the practice of the profession.

IEEE/ACM SE Code of Ethics

Some Details of the Long Version (1 of 9)

1. Public Interest

- Accept responsibility for your work
- Approve software only if you have a well-founded belief that it is
  - Safe
  - Meets specs
  - Passes its tests
  - Does not diminish quality of life
  - Does not harm privacy
  - Does not harm the environment

IEEE/ACM SE Code of Ethics

Some Details of the Long Version (2 of 9)

1. Public Interest continued

- Disclose any actual or potential danger
- Co-operate to address matters of public concern
- Be fair and avoid deception
- Consider issues that limit access to software
  - Disabilities
  - Allocation of resources
  - Economic disadvantage
- Volunteer for good causes
  - In particular, public education about the discipline

IEEE/ACM SE Code of Ethics

Some Details of the Long Version (3 of 9)

2. Client and employer

- Provide service in your area of competence
  - Disclose limitations of your education or experience
- Do not use software obtained illegally or unethically
- Use client's or employer's facilities only as authorized
- Respect privacy and confidentiality
  - Except where this violates the public interest of law
- Identify and report when a project is likely to fail, to prove too expensive or have other problems
- Avoid conflict of interest

IEEE/ACM SE Code of Ethics

Some Details of the Long Version (4 of 9)

3. Product

- Strive for achievable goals, high quality, acceptable cost and reasonable schedule
  - Ensure everyone understands the tradeoffs
  - Use quantitative estimates and state the level of uncertainty
- Use appropriate methods and standards
  - Depart from them only when ethically or technically justified
- Ensure requirements are clear and meet the user's needs
- Ensure adequate testing
- Document decisions
- Maintain the integrity of data
- Treat maintenance with the same professionalism as new development

IEEE/ACM SE Code of Ethics

Some Details of the Long Version (5 of 9)

4. Judgment

- Temper technical judgment by the need to support and maintain human values
- Only endorse items
  - you have supervised
  - you agree with
  - and in your area of competence
- Maintain professional objectivity
  - E.g. avoid promoting bad ideas to please others
- Avoid deceptive financial practices
- Avoid associating with anybody that is in a conflict of interest

IEEE/ACM SE Code of Ethics

Some Details of the Long Version (6 of 9)

5. Management

- Ensure software engineers are informed of standards before being held to them
- Ensure everybody knows policies and procedures for such things as security and privacy
- Be fair in assigning work
  - Assign work accounting for the person's level of education and experience, as well as their need to further this
- Ensure everybody knows the conditions of employment
- Offer fair and just remuneration
- Do not ask anyone to do anything inconsistent with this code
- Do not punish anyone for expressing ethical concerns.

IEEE/ACM SE Code of Ethics

Some Details of the Long Version (7 of 9)

6. Profession

- Promote public knowledge of software engineering
- Participate, as appropriate, in professional organizations
- Support others in following this code
- Obey all laws unless, in exceptional circumstances, they are inconsistent with the public interest
- Where reasonable, express concerns to people breaking this code
  - Otherwise report violations of the code to authorities

IEEE/ACM SE Code of Ethics

Some Details of the Long Version (8 of 9)

7. Colleagues

- Assist colleagues in professional development
- Credit the work of others and refrain from taking undue credit
- Review the work of others in a fair way
- Listen to the opinions, concerns or complaints of others
- Do not interfere in the career of others unless concern for the employer, client or public interest suggests otherwise
- Call on other professionals in areas outside your own competence

Engineering and Science

- The Online Ethics Center (**www.onlineethics.org**) for engineering and science seeks submissions of high-quality ethics education resources in science and engineering for inclusion in its collection. Interdisciplinary materials are particularly welcome, as are resources that promote active learning at the undergraduate or graduate level. Materials are also needed to assist faculty new to teaching in science and engineering ethics.
- You may visit the above online resource and in particular read the "Resources" section where they will come across hundreds of case studies, essays, codes and policies, multimedia and educational activities, and subject aides.
- Students may work in teams to develop various tasks related to the above activities for submission to the online resource according to the class instructor's guidance. For resource submissions, see the "Submit Materials" page.

A Project Idea:

Ethical Energy

- Identify an ethical issue of interest with future energy systems. Explore and explain different views on the above matter. Ethical perspectives employed in contest range from traditional ethics (which considers whether actions are required, recommended, or forbidden) to issues of individual (micro-ethical) and collective (macro-ethical) responsibility. The contest may focus throughout on responsibilities rather than outcomes. This ethical development task may include allocations of cost and benefits, risks and rewards, social and environmental justice, professional and organizational ethics.
- **In this task**, collaborate with peers or you may work with others outside the class to narrow down the objectives of the task. You may access online resources to create a 3-min video that reflects the above subject.

Value Change

- Value sensitive design (VSD) is a theoretically grounded approach to the design of technology that accounts for human values in a principled and comprehensive manner.
- The goal of VSD is to develop technologies that respect the values of all those impacted by them by identifying and addressing those values early in the development process, thereby limiting or eliminating potential problems once the technologies have been deployed.
  - https://delftdesignforvalues.nl/2023/how-to-deal-with-value-change-as-a-designer-engineer/